

1. DHCP Messages are sent over UDP.

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User Datagram Protocol, Src Port: 68, Dst Port: 67
  Source Port: 68
  Destination Port: 67
  Length: 310
  Checksum: 0x7896 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 83]
  [Stream Packet Number: 1]
  > [Timestamps]
  UDP payload (302 bytes)

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2. I am not sure how to draw this.

3. F8:63:3f:17:0a:7a

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Ethernet II, Src: Intel_17:0a:7a (f8:63:3f:17:0a:7a), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  Destination: Broadcast (ff:ff:ff:ff:ff:ff)
  Source: Intel_17:0a:7a (f8:63:3f:17:0a:7a)

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4. Message Type & Server Identifier Field

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Client hardware address padding: 0000000000000000
Server host name not given
Boot file name not given
Magic cookie: DHCP
  > Option: (53) DHCP Message Type (Discover)
  > Option: (61) Client identifier
  > Option: (50) Requested IP Address (192.168.0.14)
  > Option: (12) Host Name
  > Option: (60) Vendor class identifier
  > Option: (55) Parameter Request List
  > Option: (255) End
0000 ff ff ff ff ff ff f8 63 3f 17 0a 7a 08 00 45 00 .....c ?-z-E-
0010 01 4a 53 97 00 00 80 11 e6 0c 00 00 00 00 ff ff ..JS.....
0020 ff ff 00 44 00 00 43 01 36 78 96 01 01 06 00 28 76 ...D.C.6 x....(v
0030 3d f7 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....c ?-z-
0040 00 00 00 00 00 00 f8 63 3f 17 0a 7a 00 00 00 00 .....c ?-z-
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

Boot file name not given
Magic cookie: DHCP
  > Option: (53) DHCP Message Type (Request)
  > Option: (61) Client identifier
  > Option: (50) Requested IP Address (192.168.0.14)
  > Option: (54) DHCP Server Identifier (192.168.0.1)
  > Option: (12) Host Name
  > Option: (81) Client Fully Qualified Domain Name
  > Option: (60) Vendor class identifier
  > Option: (55) Parameter Request List
  > Option: (255) End
0000 ff ff ff ff ff ff f8 63 3f 17 0a 7a 08 00 45 00 .....c ?-z-E-
0010 01 64 53 98 00 00 80 11 e5 f1 00 00 00 00 ff ff ..dS.....
0020 ff ff 00 44 00 00 43 01 36 f6 8a 01 01 06 00 28 76 ...D.C.P .....(v
0030 3d f7 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....c ?-z-
0040 00 00 00 00 00 00 f8 63 3f 17 0a 7a 00 00 00 00 .....c ?-z-
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

```

a.

5. Discover/Offer/Request/ACK DHCP TxID: 0x28763df7, 0xda5c62d1

66	9.880924	192.168.0.14	192.168.0.1	DHCP	342	DHCP Release	- Transaction ID 0x23a945dc
274	15.046043	0.0.0.0	255.255.255.255	DHCP	344	DHCP Discover	- Transaction ID 0x28763df7
275	15.074692	192.168.0.1	192.168.0.14	DHCP	590	DHCP Offer	- Transaction ID 0x28763df7
276	15.076883	0.0.0.0	255.255.255.255	DHCP	370	DHCP Request	- Transaction ID 0x28763df7
277	15.134586	192.168.0.1	192.168.0.14	DHCP	590	DHCP ACK	- Transaction ID 0x28763df7
944	21.421617	192.168.0.14	192.168.0.1	DHCP	358	DHCP Request	- Transaction ID 0xda5c62d1
969	21.475096	192.168.0.1	192.168.0.14	DHCP	590	DHCP ACK	- Transaction ID 0xda5c62d1
2175	32.767034	192.168.0.14	192.168.0.1	DHCP	342	DHCP Release	- Transaction ID 0x9d4e2340
2319	37.126184	0.0.0.0	255.255.255.255	DHCP	344	DHCP Discover	- Transaction ID 0xe48e9756
2320	37.262562	192.168.0.1	192.168.0.14	DHCP	590	DHCP Offer	- Transaction ID 0xe48e9756
2323	37.265111	0.0.0.0	255.255.255.255	DHCP	370	DHCP Request	- Transaction ID 0xe48e9756
2324	37.305000	192.168.0.1	192.168.0.14	DHCP	590	DHCP ACK	- Transaction ID 0xe48e9756

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6. IP Addresses:

274	15.046043	0.0.0.0	255.255.255.255	DHCP	344	DHCP Discover	- Transaction ID 0x28763df7
275	15.074692	192.168.0.1	192.168.0.14	DHCP	590	DHCP Offer	- Transaction ID 0x28763df7
276	15.076883	0.0.0.0	255.255.255.255	DHCP	370	DHCP Request	- Transaction ID 0x28763df7
277	15.134586	192.168.0.1	192.168.0.14	DHCP	590	DHCP ACK	- Transaction ID 0x28763df7

- a. Discover:

- i. 0.0.0.0/255.255.255.255

- b. Offer:

- i. 192.168.0.14/255.255.255.255

- c. Request:

- i. 0.0.0.0/255.255.255.255

- d. ACK:

- i. 192.168.0.14/255.255.255.255

7. DHCP Server IP: 192.168.0.14

✓ Option: (54) DHCP Server Identifier (192.168.0.1)
Length: 4

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8. IP Address Offered: 192.168.0.14, Offer Message contains DHCP Address Offered

✓ Dynamic Host Configuration Protocol (Offer)

Message type: Boot Reply (2)
Hardware type: Ethernet (0x01)
Hardware address length: 6
Hops: 0
Transaction ID: 0x28763df7
Seconds elapsed: 0
➤ Bootp flags: 0x0000 (Unicast)
Client IP address: 0.0.0.0
Your (client) IP address: 192.168.0.14
Next server IP address: 192.168.0.1
Relay agent IP address: 0.0.0.0
Client MAC address: Intel_17:0a:7a (f8:63:3f:17:0a:7a)
Client hardware address padding: 00000000000000000000
Server host name not given
Boot file name not given
Magic cookie: DHCP

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9. The value that indicates there is no relay agent is 0.0.0.0 and I also have a value for the relay agent of 0.0.0.0 indicating that I too did not have a relay agent.

Relay agent IP address: 0.0.0.0

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10. Purpose of Router + Subnet Mask

- The subnet mask line tells the client which subnet mask to use
- The router line indicates where the client should send messages by default

11. The client accepts the IP address given in the offer message within the request message. After being offered the IP address 192.168.0.14 in the offer message, my client sent back a message further requesting that specific IP address.

12. Lease Time / Total Lease Time

- The purpose of lease time is to tell the client how long they can use the specific IP address assigned by the server before they will have to be assigned a new one.
- The lease time in my experiment is 1 hour.

✓ Option: (51) IP Address Lease Time
Length: 4
IP Address Lease Time: 1 hour (3600)

13. Purpose of DHCP Release Message / Ack / Consequence if Lost

- The purpose of the release message is to release the IP address back to the server.
- There is no verification that the release message has been received by the server.
- If the message is lost, the client releases the IP address, but the server will not reassign that address until the clients lease on the address expires.

14. Yes there are a lot; they appear to be broadcasts sent out by the network to build up the known IP addresses by the clients network.

60	8.034668	Intel_17:0a:7a	Commscope_a7:40:a7	ARP	42	192.168.0.14 is at f8:63:3f:17:0a:7a
315	15.270705	Intel_17:0a:7a	Broadcast	ARP	42	Who has 192.168.0.1? Tell 192.168.0.14
316	15.297807	Commscope_a7:40:a7	Intel_17:0a:7a	ARP	56	192.168.0.1 is at 10:86:8c:a7:40:a7
317	15.300900	Intel_17:0a:7a	Broadcast	ARP	42	Who has 192.168.0.14? (ARP Probe)
318	15.300992	Intel_17:0a:7a	Roku_2a:b9:05	ARP	42	Who has 192.168.0.5? Tell 192.168.0.14
319	15.301018	Intel_17:0a:7a	Roku_d6:c5:55	ARP	42	Who has 192.168.0.2? Tell 192.168.0.14
324	15.303476	Roku_d6:c5:55	Intel_17:0a:7a	ARP	42	192.168.0.2 is at 08:05:81:d6:c5:55
325	15.304292	Roku_2a:b9:05	Intel_17:0a:7a	ARP	42	192.168.0.5 is at b8:a1:75:2a:b9:05
349	15.442725	Intel_17:0a:7a	Broadcast	ARP	42	Who has 192.168.0.1? Tell 192.168.0.14
350	15.451912	Commscope_a7:40:a7	Intel_17:0a:7a	ARP	56	192.168.0.1 is at 10:86:8c:a7:40:a7
399	16.299927	Intel_17:0a:7a	Broadcast	ARP	42	Who has 192.168.0.14? (ARP Probe)
436	16.798650	Intel_17:0a:7a	Broadcast	ARP	42	Who has 169.254.49.213? (ARP Probe)
474	17.291439	Intel_17:0a:7a	Broadcast	ARP	42	Who has 192.168.0.14? (ARP Probe)
578	17.801566	Intel_17:0a:7a	Broadcast	ARP	42	Who has 169.254.49.213? (ARP Probe)
630	18.299582	Intel_17:0a:7a	Broadcast	ARP	42	ARP Announcement for 192.168.0.14
1366	22.193166	Intel_17:0a:7a	Broadcast	ARP	42	Who has 192.168.0.1? Tell 192.168.0.14
1370	22.203075	Commscope_a7:40:a7	Intel_17:0a:7a	ARP	56	192.168.0.1 is at 10:86:8c:a7:40:a7
1383	22.234111	Intel_17:0a:7a	Broadcast	ARP	42	Who has 192.168.0.1? Tell 192.168.0.14
1386	22.237470	Commscope_a7:40:a7	Intel_17:0a:7a	ARP	56	192.168.0.1 is at 10:86:8c:a7:40:a7
2337	37.527150	Intel_17:0a:7a	Broadcast	ARP	42	Who has 192.168.0.1? Tell 192.168.0.14
2367	37.611861	Commscope_a7:40:a7	Intel_17:0a:7a	ARP	56	192.168.0.1 is at 10:86:8c:a7:40:a7
2376	37.717484	Intel_17:0a:7a	Broadcast	ARP	42	Who has 192.168.0.1? Tell 192.168.0.14
2377	37.732117	Commscope_a7:40:a7	Intel_17:0a:7a	ARP	56	192.168.0.1 is at 10:86:8c:a7:40:a7
2378	37.797407	Intel_17:0a:7a	Broadcast	ARP	42	Who has 192.168.0.14? (ARP Probe)